A/B Test Analysis: Free Trial Screener

1. CONTEXT

A SaaS learning platform tested a pre-enrollment screener shown to users after clicking Start Free Trial. Users who indicated fewer than 5 hours/week of availability were redirected to free course materials. The control group saw no screener. The experiment ran for 37 days across ~690,000 pageviews split equally between groups.

2. HYPOTHESIS

H0 (null): The screener has no effect on net conversion rate (Payments / Clicks).
H1 (alternative): The screener increases net conversion by filtering out low-commitment users.

Primary metric: Net Conversion (Payments / Clicks) practical threshold: +/-0.75pp
Guardrail metric: Gross Conversion (Enrollments / Clicks) practical threshold: +/-1.00pp

3. METHOD

Two-proportion z-test (two-tailed, alpha = 0.05). Metrics computed over the 23 days with complete enrollment and payment data (~18,000 clicks per group). Sanity checks confirmed a clean 50/50 traffic split (p = 0.29) and equal click-through rates across groups (p = 0.93), validating the randomization.

4. RESULTS

Metric	Control	Experiment	Delta	p-value	Decision
Gross Conversion	21.9%	19.8%	-2.06pp	<0.001	Sig. drop
Net Conversion	11.8%	11.3%	-0.49pp	0.156	Not sig.

95% CI Gross Conversion: [-2.91pp, -1.20pp] | 95% CI Net Conversion: [-1.16pp, +0.19pp]

5. RECOMMENDATION: DO NOT LAUNCH

The screener reduced gross enrollment by a statistically and practically significant margin (-2.06pp, p<0.001), confirming it filtered users as designed. However, net conversion did not improve (-0.49pp, p=0.156). The confidence interval for net conversion [-1.16pp, +0.19pp] cannot rule out a negative revenue impact.

Launching would reduce enrollment without a compensating gain in payment quality -- a worse expected outcome than the baseline.

Suggested follow-up experiments:
- Lower the commitment threshold from 5 to 3 hours and re-test.
- Test the messaging shown to screened-out users to recover lost conversions.
- Segment results by traffic source (paid vs organic) to find where the screener may work.

Gnanendra Reddy | github.com/gnanendrart | linkedin.com/in/gnanendrart